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Convolutional neural network framework for EEG-based ADHD diagnosis in children

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Abstract

Purpose: Attention-deficit hyperactivity disorder (ADHD) stands as a significant psychiatric and neuro-developmental disorder with global prevalence. The prevalence of ADHD among school children in India is estimated to range from 5% to 8%. However, certain studies have reported higher prevalence rates, reaching as high as 11%. Utilizing electroencephalography (EEG) signals for the early detection and classification of ADHD in children is crucial.

Methods: In this study, we introduce a CNN architecture characterized by its simplicity, comprising solely two convolutional layers. Our approach involves pre-processing EEG signals through a band-pass filter and segmenting them into 5-s frames. Following this, the frames undergo normalization and canonical correlation analysis. Subsequently, the proposed CNN architecture is employed for training and testing purposes.

Results: Our methodology yields remarkable results, with 100% accuracy, sensitivity, and specificity when utilizing the complete 19-channel EEG signals for diagnosing ADHD in children. However, employing the entire set of EEG channels presents challenges related to the computational complexity. Therefore, we investigate the feasibility of using only frontal brain EEG channels for ADHD detection, which yields an accuracy of 99.08%.

Conclusions: The proposed method yields high accuracy and is easy to implement, hence, it has the potential for widespread practical deployment to diagnose ADHD.

Keywords: ADHD, Canonical correlation analysis, CNN, EEG, Frontal brain

Introduction

Attention-deficit hyperactivity disorder (ADHD) is a neuro-developmental disorder in children, marked by symptoms such as inattention, excessive activity, and impulsivity, leading to challenges in behavioral, psychological, academic, and social aspects throughout their life [1]. Individuals with ADHD often display symptoms such as forgetfulness, disorganization, and difficulties maintaining concentration and attention [2]. Timely diagnosis and intervention can facilitate the restoration of neuronal connections to the prefrontal cortex, enabling individuals to return to normal daily activities [3]. Failure to address ADHD at an early stage may lead to persistent symptoms into adulthood, increasing the risk of antisocial behaviors and depression. Other undesirable outcomes may include involvement in crime, academic

underachievement, interpersonal relationship difficulties, and reduced employability. A prompt and accurate diagnosis of ADHD holds significant value for advancing clinical studies aimed at discovering more effective treatments for this condition.

Researchers have explored diverse neuro-imaging techniques and analytical approaches over the years to enhance the understanding of ADHD. According to the 2016 statistics from the Center for Disease Control and Prevention (CDC), more than 6.1 million children between the ages of 4 and 17 have received a diagnosis of ADHD [4]. In India, a study conducted in Coimbatore [5] revealed a higher prevalence of 11.32% for ADHD in children, compared to the global estimate. It highlighted peak prevalence rates at ages 9 (26.4%) and 10 (25%). Moreover, the research indicated a higher occurrence of ADHD in males, accounting for 66.7% of the cases. The study observed that children with ADHD not only exhibited challenges in academic performance and behavior, but

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also encountered difficulties related to reading and writing. The advancement of objective diagnostic methods, particularly those rooted in biological signals that mirror ADHD behaviors and its sub-types, would bring significant benefits to children, parents, and community health. Presently, the examination and diagnosis of ADHD rely on time-consuming traditional methods, including clinical interviews, observations, and ratings obtained from various sources such as parents and teachers [6]. These conventional evaluation procedures are susceptible to ambiguity [7]. Hence, there is a pressing need for more efficient and objective diagnostic approaches considering biological signals to enhance the accuracy and expediency of ADHD assessments.

Electroencephalography (EEG) is a readily accessible and cost-effective neuro-imaging method that enables a comprehensive representation of various brain processes [8]. Techniques focusing on brain connectivity offer a systemic approach to understanding the pathophysiology of neuro-psychiatric disorders. Examining the atypical patterns within the entire brain system proves valuable in gaining insights into neuro-developmental disorders, given the crucial role of coherent interaction among brain regions in human cognition and behavior [9–12]. The analysis of connectivity between brain regions primarily involves two approaches: functional connectivity and effective connectivity. Functional connection refers to the statistical interdependence of activities in several brain parts, whereas effective connectivity emphasizes the influence exerted by one part over another [13]. Authors in [7] introduced a method that uses EEG signals for the detection of ADHD, wherein tunable Q-wavelet transform (TQWT) is applied to extract nonlinear features from the input signals and an artificial neural network (ANN) is employed as a classifier. Ahmadlou et al. [14] proposed a technique for wavelet synchronization of EEG signals to extract precise features within five sub-bands. These features are then utilized in conjunction with a radial basis function (RBF) neural network classifier to distinguish between healthy individuals and those suffering with ADHD.

In the study conducted by Allahverdy et al. [15], visual attention tasks are employed to identify ADHD amongst 49 children aged between 7 to 12 years, comprising a group of 20 healthy children and 29 children diagnosed with ADHD. The nonlinear features, including Higuchi fractal dimension, Lyapunov exponent, Sevcik fractal dimension, and Katz fractal dimension are extracted from the EEG data captured from the frontal lobe electrodes, and are analyzed using a multi-layer perceptron (MLP) neural network. Further, authors in [16] introduced an EEG-centered deep learning framework called ADHD/CD-NET, designed to effectively differentiate between

ADHD, and normal children. The process involves initial segmentation of the EEG signals, followed by their transformation into continuous wavelet transform (CWT) correlation matrices on a per-channel basis. Subsequently, these matrices serve as the training input for the convolutional neural network (CNN) model. A novel ternary motif pattern (TMP) has been developed in [17], adopting a hand-modeled classification approach inspired by deep learning networks. The TQWT has been employed to produce wavelet sub-groups. The proposed TMP has been applied in combination with statistical measures to derive information from the raw EEG signals as well as the wavelet bands generated by TQWT. The model is referred as TMP19, wherein the most informative features are chosen using the neighborhood component analysis (NCA), and the classification is performed by the k-nearest neighbor (kNN) classifier.

Authors in [18] introduced a feature extraction method that assesses connectivity tensors amongst EEG channels to construct the input structure for the classification model. These tensors capture correlations between EEG channels across various time frames, maintaining temporal as well as spatial structures of the input data and lower the input dimensions. They utilized a neural network, convolutional long short-term memory (ConvLSTM) along with an attention mechanism to distinguish between ADHD patients and the normal group. On the other hand, nonlinear causal relationship estimation by ANN (nCREANN) technique is applied in [13] to analyze the connectivity of brain regions in children with ADHD. The EEG signals collected during a visual attention exercise are analyzed and compared to those from age-matched typically-developing (TD) volunteers. Analysis of these existing schemes indicates that some researchers have delved into the application of deep learning algorithms for screening ADHD based on EEG signals, while others have investigated more streamlined machine learning methodologies. A majority of these studies have focused on extracting nonlinear features and employing standard classifiers like support vector machine (SVM), MLP, and kNN for the classification process [19–22]. Such hand-crafted features need to be meticulously engineered before being fed into a classifier. The performance of their approach relies on identifying and crafting features that are relevant for the classification task. On the contrary, some other studies utilizing deep learning networks for the detection of ADHD, convert the EEG signals into images and pre-process them before providing as input to the deep network.

In this work, we are proposing a novel methodology to detect ADHD. The raw EEG signals are filtered, followed by segmentation and normalization. Thereafter, canonical correlation analysis (CCA) is performed, followed by

training and testing using the proposed CNN. The key features of this method are as follows:

- The CNN architecture comprises only two convolutional layers, hence, the associated hardware and computational expenses are minimal.
- Raw EEG signals are considered, rather than converting them to images.
- Proposed network can extract relevant features directly from the data, instead of relying on manually crafted features.
- This model achieves an accuracy, sensitivity and specificity of 100%, which is higher than existing state-of-the-art methods.
- Remarkable accuracy of 99.08% is achieved using an optimal selection of 5 channels instead of the full 19 channel data.

The rest of the paper follows this structure: “**Materials and methods**” section provides an overview of the materials and method utilized. “**Results and discussions**” section presents the achieved outcomes and subsequent discussion, and lastly, “**Conclusions and future work**” section, summarizes the conclusions drawn from this work and suggests avenues for further research.

Materials and methods

The schematic representation of the proposed approach is depicted in Fig. 1, highlighting the sequential procedures adopted in this work. The raw EEG signals undergo pre-processing steps, including band-pass filtering after which the signals are segmented into frames of 5-s. These frames are then normalized and CCA is performed on the normalized frames, before being fed to the proposed CNN model.

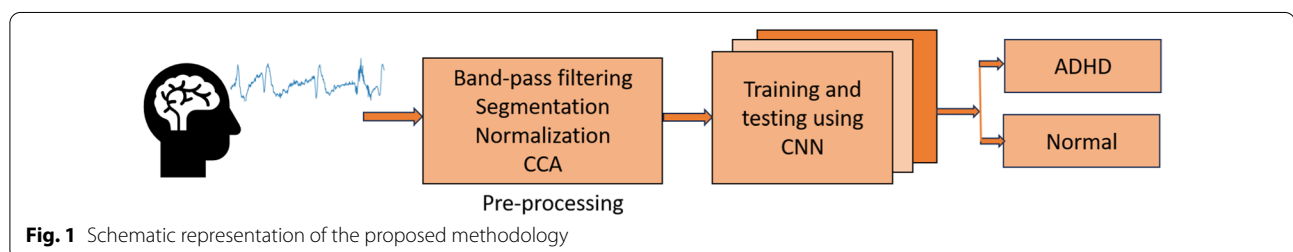
Dataset

The dataset for this study is sourced from a study conducted by Nasrabadi et al. [23]. The dataset comprises EEG records of 61 children diagnosed with ADHD, and 60 healthy controls (HC), all aged between 7 and 12 years. ADHD diagnosis was performed by a qualified

psychiatrist who followed DSM-IV criteria, and ADHD patients were given Ritalin for a maximum of 6 months. The HC group consisted of children without psychiatric illnesses, epilepsy, or high-risk behaviors. The EEG recordings were obtained during a visual attention test, where children counted cartoon characters in presented images. This data was collected at the Psychology and Psychiatry Research Center at Roozbeh Hospital in Tehran, Iran, utilizing 19 channels (Fz, Cz, Pz, C3, T3, C4, T4, Fp1, Fp2, F3, F4, F7, F8, P3, P4, T5, T6, O1, O2), sampled at a frequency of 128 Hz.

Pre-processing

To prepare the EEG signals for analysis, it is essential to address various artifacts and noises inherent in the data. Due to the high temporal resolution inherent in EEG signals, the recording process may be susceptible to artifacts. The data can be affected by both internal and external artifacts, with external artifacts arising from the experimental errors and environmental noise. Employing filtering techniques proves effective in mitigating the impact of external artifacts on the data [24, 25]. Internal artifacts, referred to as physiological artifacts, have the potential to degrade the quality of EEG signal recordings. In numerous studies, a common practice involves manual pre-processing to eliminate these internal artifacts. In this work, a band-pass filter of order 4 is used having cut-off frequencies of 2 Hz and 47 Hz to eliminate the external artifacts. This filtering is crucial for removing unwanted frequency components from the EEG signals. External artifacts such as power line noise (50/60 Hz) and other low-frequency drifts are effectively eliminated. The chosen frequency range captures the most relevant EEG activity crucial for ADHD detection, since ADHD is mostly detected by theta, beta and alpha bands that are included in the chosen range [26, 27]. Thereafter, frames are generated using segmentation of EEG records. The segmentation allows for the analysis of shorter, more manageable time intervals. This is essential for identifying transient events and patterns associated with ADHD. Moreover, breaking down the EEG data into multiple frames provides a larger dataset for statistical analysis,



enhancing the reliability of the results. After segmentation, normalization is performed to ensure that the EEG signals from different frames and subjects are on the same scale. This is crucial for reducing variability and improving the consistency of the analysis. By normalizing the data, the subsequent feature extraction and classification steps become more accurate, as the influence of varying amplitudes is minimized. Lastly, we have employed CCA which is a powerful statistical technique used to remove internal artifacts such as eye blinks and muscle movements, by identifying and separating the correlated noise components from the EEG signals. By reducing these artifacts, CCA helps in preserving the true neural activity, leading to a more accurate representation of the underlying brain function.

Canonical correlation analysis (CCA)

It is a multivariate statistical technique that examines the links between two sets of data. The goal is to identify linear combinations of variables from each set, known as canonical variables, such that the correlation between these canonical variables is maximized. CCA is widely used to investigate the relationship between two sets of variables and unveil the inherent patterns that exist between them [28]. Let A and B denote two sets of variables, containing m and n variables, respectively, measured across z observations. The canonical correlation analysis seeks to find linear combinations of P and Q from the sets A and B , respectively, such that the correlation between P and Q is maximized. Consider $A = [A_1, A_2, \dots, A_m]$ to be the matrix of variables in set A and correspondingly, set $B = [B_1, B_2, \dots, B_n]$. The mean, variance and co-variance of A and B are represented as: $\mu_a, \mu_b, \text{Cov}(A), \text{Cov}(B)$, and $\text{Cov}(A, B)$, respectively. Also $B^t A = [A^t B]^t$ is a symmetric matrix of dimension $(n \times m)$. Let us assume P and Q are linear combinations and can be calculated as: $P = c^t A, Q = d^t B$, where c and d are the vector coefficients for the linear combinations and the correlation between each pair of linear combinations P and Q is of maximum value, i.e.,

$$\text{maximumCorrelation}(P, Q) = \rho_1 \quad (1)$$

The correlation coefficient (CC) between P and Q is known as the canonical correlation and mathematically is equal to:

$$\text{CC}(P, Q) = \frac{c^t \text{Cov}(A, B) d}{\sqrt{(c^t \text{Cov}(A) c)(d^t \text{Cov}(B) d)}} \quad (2)$$

In this work, CCA is applied to detect similar patterns or correlations between EEG data. By applying 19-channel EEG data, $l = \min(m, n)$ canonical variates that are highly connected are obtained, i.e., it finds l canonical axis

in m dimensional A and in n dimensional B , such that projections of data onto each pair of canonical variables are maximally correlated. Subsequently, a subset of these canonical pairs is selected to retain the most pertinent information. Following this selection, the transformed data is projected onto the chosen canonical variables. Figure 2 depicts a segment of EEG signal, before and after CCA is applied.

Proposed CNN architecture

CNNs consist of convolutional layers that operate by extracting features from the input. The features obtained are then processed through dense layers to make predictions. Convolution and sub-sampling techniques are employed across multiple layers to find significant features from input data. Generally, CNN architectures include numerous hidden layers [29]. In this work, a lightweight CNN architecture is designed for ADHD detection that utilizes EEG signals. A notable advantage of this architecture is its simple design. The EEG signals employed in this investigation are 19-channel signals, with a sampling frequency of $f_s = 128$ Hz. To train the model, we opted for the Adam optimizer and utilized a binary cross-entropy loss function. Figure 3 provides a visual representation of the proposed architecture. The kernel size ' k ' for the first layer is chosen to be 64, while for the separable convolution layer, $k = 16$.

The suggested framework comprises two convolutional layers crafted for the proficient examination of EEG signals.

The first layer is a temporal convolutional layer, followed by a separable convolutional layer. To capture relevant information from EEG signals ranging from 2 to 47 Hz, kernel size of 64 is used for the first convolutional layer. It helps to extract data above 2 Hz effectively. This layer uses flexible filters (kernels) on input, moving methodically across it, to generate feature maps through element-wise multiplications and summations. The output value o_{ij} of an individual neuron in a convolutional layer is expressed as:

$$o_{ij} = f \left(\sum_{m=0}^{M-1} \sum_{n=0}^{N-1} x_{i+m, j+n} \cdot w_{mn} + b \right), \quad (3)$$

where $f(\cdot)$ represents the activation function employed, $x_{i+m, j+n}$ denotes the input value at coordinates $(i + m, j + n)$ in the input, while w_{mn} is the weight of the filter at coordinates (m, n) and b is the bias. To decrease the spatial dimensions of the data while preserving the essential information, we use average pooling, with a pool size of 3. Max pooling preserves the highest value within a localized area, i.e.,

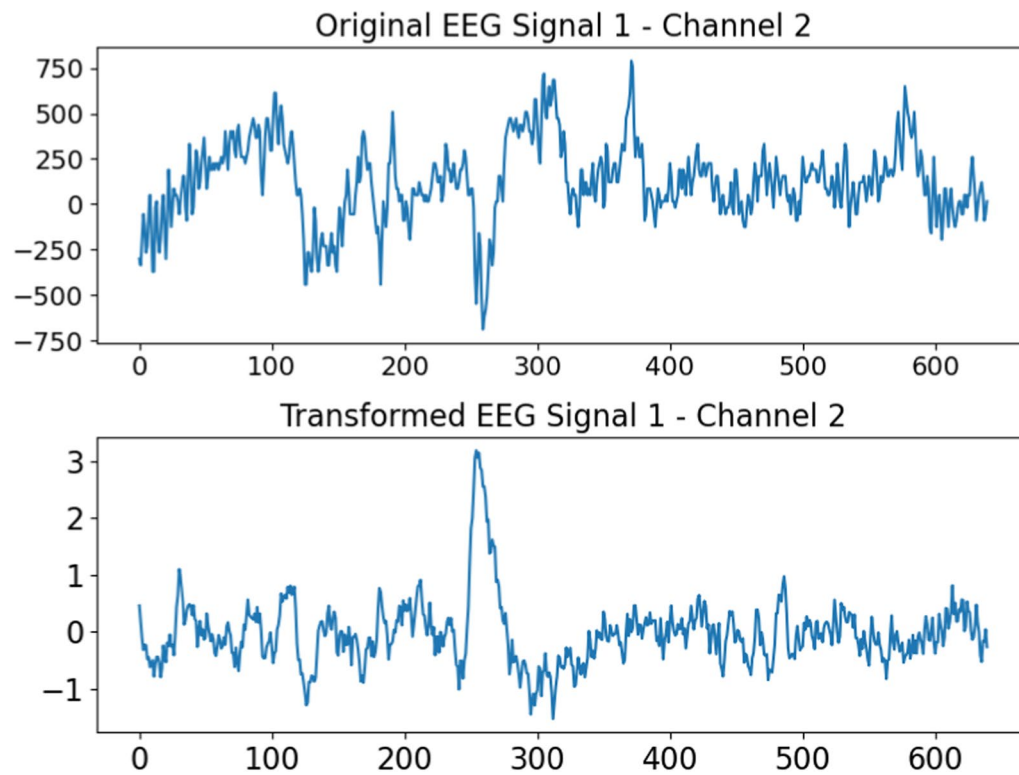


Fig. 2 A segment of EEG signal before and after application of CCA

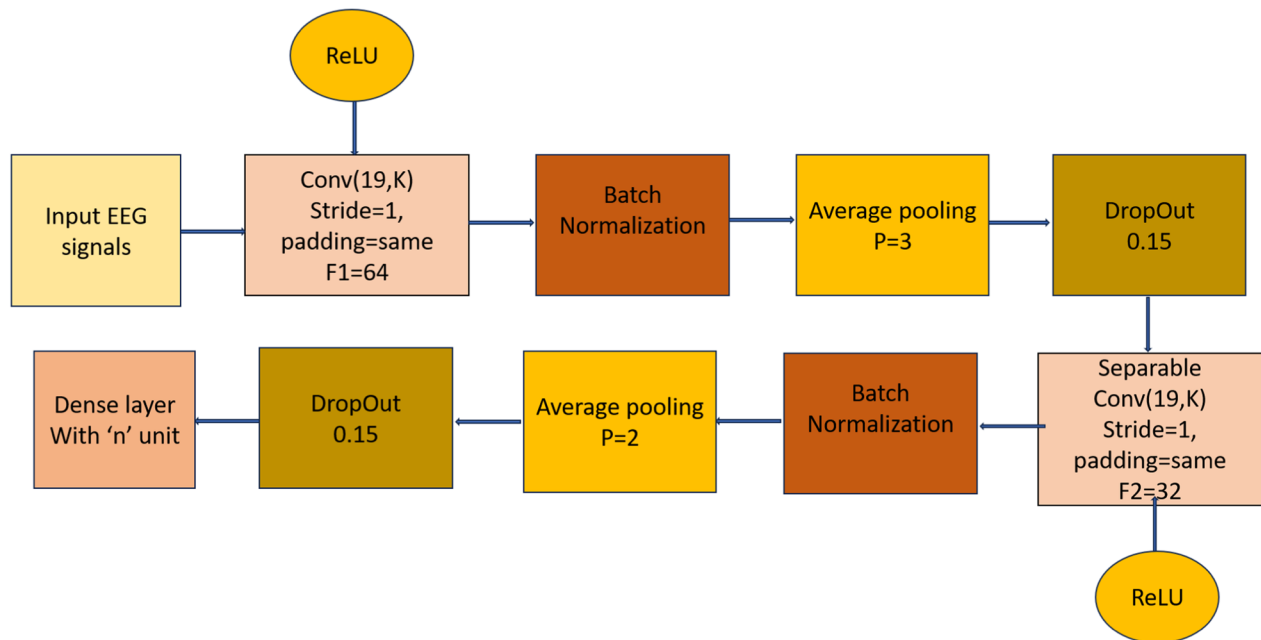


Fig. 3 Architecture of proposed CNN architecture

$$y_{ij} = \max_{m=0}^{M-1} \max_{n=0}^{N-1} x_{i+m,j+n}, \quad (4)$$

where y_{ij} represents the output in the pooled feature map.

For the second convolutional layer, separable convolution with 32 filters and a kernel size of 16 is employed. This layer is a variant of the standard convolutional layer, designed to minimize the number of parameters and computational complexity, while maintaining or enhancing the representation capability. Thereafter, average pooling is employed again to down-sample the output of the second layer by a factor of 2. For both the convolutional layers, rectified linear unit (ReLU) is used as the activation function. It is defined as:

$$f(x) = \max(0, x), \quad (5)$$

where $f(x)$ is the output and x is the input value to the function. Batch normalization is also applied after each convolutional layer, which stabilizes training and improves performance. Subsequently, fully connected layers establish connections between neurons for classification. The neuron's output in the fully connected layer can be expressed as:

$$O_i = f\left(\sum_{j=0}^{J-1} y_j \cdot w_{ji} + b_i\right) \quad (6)$$

This final layer classifies the signals into two classes: ADHD and normal. For the output layer, the sigmoid activation function is used. It generates an S-shaped curve, gradually transitioning the output from 0 to 1. To prevent over-fitting and enhance generalization, we employ dropout with a rate of 0.15. A summary of the different layers in the proposed CNN architecture is provided in Table 1, along with the necessary characteristics. Further, the values for other parameters utilized in this work are shown in Table 2, where LR denotes the learning rate.

Table 1 Architecture of the proposed CNN model

Layer	Filter (F)	Kernel size	Activation
Input		(C,T)	
Conv	($F_1 = 64$)	(1,64)	ReLU
Batch normalisation			
Average pool		(1,3)	
Separable conv	($F_1 = 32$)	(1,16)	ReLU
Batch normalization			
Average pool		(1,2)	
Drop out			
Dense			sigmoid

Table 2 Parameters for the proposed architecture

Parameter	Value
LR	0.0001
Dropout	0.15
Iterations	100
Optimizer	Adam
Batch size	64

Evaluation metrics

To ensure that the proposed model is evaluated accurately, we utilize 5-fold cross-validation scheme, which involves dividing our dataset into five equal parts, or 'folds.' Each fold contains a representative mix of samples from all classes, mirroring the distribution of the original dataset. During each iteration of cross-validation, a single fold is set aside for testing, while the rest four folds are allocated for training the model and fine-tuning the hyper-parameters. This process is repeated five times, with each fold serving as the test set once, and the results are averaged across these five iterations. By testing and training on different parts of the dataset in each iteration, we obtain a stable and reliable estimate of the model's performance across the entire dataset, minimizing the risk of bias compared to a single train-test split.

To quantify the performance of the classification algorithm, we use three metrics: accuracy, sensitivity, and specificity. They help us in assessing the effectiveness of the proposed model for detecting ADHD from EEG signals. Each metric captures different aspects of the classification task. While accuracy provides an overall view of the model performance, it can be misleading in imbalanced datasets where one class is dominating. The classifier's capacity to detect positive samples accurately is reflected by sensitivity, whereas specificity gauges the ratio of true negatives amongst the actual negatives. Considering both sensitivity and specificity offers a balanced evaluation of the model's performance across both the classes. These performance metrics are derived from the confusion matrix as follows [30]:

$$\text{Accuracy (acc)} = \frac{TP + TN}{TP + TN + FP + FN} \quad (7)$$

$$\text{Sensitivity (sen)} = \frac{TP}{TP + FN} \quad (8)$$

$$\text{Specificity (spe)} = \frac{TN}{TN + FP} \quad (9)$$

Here, true positives (TP) and true negatives (TN) represent correct predictions of EEG frames as normal or

ADHD, while incorrect classifications are reflected by false positives (FP) and false negatives (FN).

Results and discussions

In this study, the proposed CNN network is implemented using the TensorFlow backend and Jupyter software. EEG signal processing has gained considerable attention in recent decades, emerging as a prominent research area. However, analyzing a vast array of channels presents various challenges, including issues like over-fitting, substantial computational demands, and time constraints [31]. Identifying the optimal channels becomes crucial for enhancing model performance while simultaneously economizing computational resources and time. The dataset utilized in this work comprises 19 channel EEG data. Several research studies propose various methods for choosing the most effective features computed across all channels. Opting for a limited set of features also implies a reduction in the number of channels utilized. The calculation of features for all channels during training, followed by an optimization process for feature selection can be computationally intensive. Thus, finding the optimal channels to detect ADHD is a better alternative.

Booth et al. [32] discovered that individuals with ADHD exhibited diminished activation across a broad network of frontal areas, particularly in the right hemisphere. Additionally, research indicates a decrease in functional connectivity within fronto-striatal and fronto-parietal regions during rest in children with ADHD [33]. Further, authors in [34] observed significant differences in the information pathways involving connections

among electrodes in the frontal region, while considering ADHD and healthy children. In view of these observations, we perform two separate experiments. Firstly, we consider the entire 19 channel EEG data. Next, we experiment with the 5 frontal channels only to reduce the dimensionality of the input data and the associated computational load. Figure 4 shows the locations for the 19 EEG channels, wherein the 5 frontal channels are highlighted in yellow. Table 3 lists the accuracy values for these individual channels: f3, f4, f7, f8, and fz. It is observed that all these channels yield an accuracy of more than 95%. Further, the optimal channels selected in [15, 35] also have these channels in common. Considering the combined EEG data for these 5 frontal channels, the proposed CNN architecture achieves a remarkable accuracy of 99.08%. The confusion matrix for this classification task is shown in Fig. 5.

The performance metrics for the proposed method are listed in Table 4. The impact of applying CCA technique is also demonstrated by comparing the results with/without CCA. It is evident that the application of CCA improves the accuracy by more than 10%. The proposed CNN architecture achieves an accuracy, sensitivity, and specificity of 100%, using all the 19 EEG channels. On the other hand, 5 channel EEG data yields accuracy, sensitivity, and specificity of 99.08%, 100% and 97.83%,

Table 3 Accuracy values for the individual channels

Channel	Accuracy (%)
f7	99.89
f8	99.78
f3	98.72
f4	95.03
fz	97.86

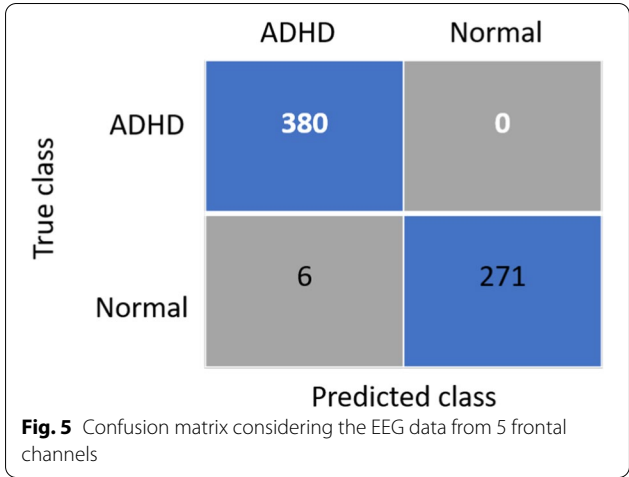
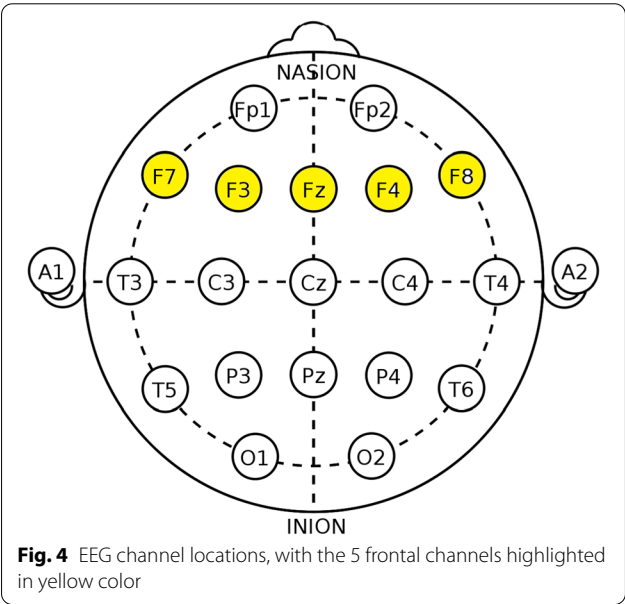


Table 4 Performance of the proposed scheme, with and without the application of CCA

Proposed system	No. of Channels	Accuracy (%)	Sensitivity (%)	Specificity (%)
With CCA	19	100.0	100.0	100.0
	5	99.08	100.0	97.83
Without CCA	19	89.93	100.0	88.67
	5	88.40	91.61	82.80

**Fig. 6** Training loss and accuracy of the proposed model**Table 5** Comparison of the proposed work with the existing methods

Study	Classifier	Accuracy (%)
Chen et al. [36]	SVM	84.6
Alkahtani et al. [37]	CNN	97.75
Parashar et al. [38]	Adaboost	84.00
Maniruzzaman et al. [35]	Gaussian process	97.5
Allahvardey et al. [15]	MLP	96.7
Ahire et al. [39]	Naive bayes	96.00
Atila et al. [40]	SVM	98.9
Taghibeyglou et al. [41]	Logistic regression	95.83
Moghaddari et al. [42]	CNN	99.06
Tosun et al. [43]	LSTM	92.15
This work	CNN	99.08

respectively. It is observed that our model can distinguish ADHD signals from normal signals quite accurately. Figure 6 illustrates the training loss and the accuracy achieved by the model in each epoch. It can be seen that the loss is converging towards zero while the accuracy gets better with each epoch.

The comparison of the proposed work with the existing state-of-the-art methods is provided in Table 5. Authors

in [36] utilized the minimum redundancy maximum relevance (mRMR)-test to extract important features from various time and frequency features. They reported an accuracy of 84.59% using SVM classifier. Alkahtani et al. [37] pre-processed raw brain signals to fetch meaningful features, followed by feature selection techniques to identify relevant variables. The selected features are processed using AdaBoost, Random Forest (RF), Decision Trees (DT), Long short-term memory (LSTM) and CNN models, wherein an accuracy of 97.75% is reported using CNN. Further, three different classifiers, i.e., AdaBoost, RF, and SVM are employed for the classification task in [38], with AdaBoost yielding an accuracy of 84.0%. Machine learning (ML) techniques are investigated in [44] by identifying the optimal channels and their significant features. Hybrid channel selection is performed using a combination of SVM and statistical *t* test, and a logistic regression model is utilized to select important features. Six different ML-classifiers are applied, including Gaussian process classification (GPC), RF, kNN, MLP, DT, and logistic regression. Authors in [15] used a MLP network that considers nonlinear features of EEG to obtain an accuracy of 96.7% using frontal lobe electrodes only. Ahire et al. [39] employed discrete wavelet transform (DWT) and extracted statistical features of the EEG signals, resulting in an accuracy of 96% with Naive Bayes (NB) classifier. On the contrary, Atila et al. [40] developed a method using patterns derived from Ulam's spiral and Sophia Germain's prime numbers. They utilized an SVM classifier in conjunction with 10-fold cross-validation strategy for detection of ADHD and achieved an accuracy of 98.9%.

A CNN architecture is combined with traditional ML models in [41], which utilized the raw EEG signals directly as input for diagnosing ADHD in children. This approach eliminated the need for signal transformation or artifact rejection techniques. Following this, feature maps from various layers of the trained CNN are extracted and employed to train classical classifiers like SVM, RF, and logistic regression, achieving a maximum accuracy of 95.83%. Moghaddari et al. [42] constructed a deep learning model utilizing a CNN, wherein the EEG signals underwent pre-processing to remove noise and artifacts. Subsequently, theta, alpha, beta, and gamma

frequency bands are extracted, resulting in the formation of a color RGB image with three channels. Finally, the generated images are fed into a 13-layer CNN for feature extraction and classification purposes, resulting in an accuracy of 99.06%. Authors in [43] make an attempt to identify the most effective channel and recording condition for diagnosing ADHD. The study applied raw data to classifiers including LSTM, SVM, and ANN. LSTM yielded the highest accuracy, reaching 88.88% on the 'Fp1, F7' channels and 92.15% during eyes-closed resting state. Spectral entropy is noted to have a positive impact on accuracy. Consequently, the study highlights the effectiveness of utilizing the potential difference between Fp1, F7 electrodes during the eyes-closed resting state for ADHD diagnosis. In comparison to these techniques, the proposed method is more accurate and efficient, yielding an accuracy of 99.08%, using only 5-channel raw EEG data as input and a small CNN architecture. Moreover, 100% accuracy is achieved by considering entire 19-channel EEG data.

The obtained accuracy and recall rates highlight the potential of proposed approach as a powerful tool in both clinical practice and research. The findings of this study can be used by clinicians to confirm or cross-validate their assessments, potentially leading to a more accurate and early diagnosis. The proposed work can complement subjective assessments like behavioral questionnaires and interviews, thereby reducing diagnostic bias and variability. Development of portable and affordable EEG devices can facilitate widespread use in clinical and non-clinical settings, making ADHD diagnostics more accessible. Moreover, changes in neural activity patterns over time can be observed to monitor the effectiveness of treatments, such as medication or behavioral therapy. The proposed methodology may be integrated into telemedicine platforms to enable remote assessment and monitoring, expanding the reach of ADHD care. For research, it opens new avenues for understanding the neurobiological basis of ADHD, developing better diagnostic algorithms, and exploring innovative applications in real-world settings. Furthermore, combining EEG with other neuroimaging techniques or physiological measures can provide a more comprehensive understanding of ADHD.

Conclusions and future work

In this work, a lightweight CNN model is proposed to assist the medical experts in an accurate detection of ADHD. The proposed architecture has only two convolutional layers making it simple and easy to implement in practical scenarios. The input to the proposed architecture is pre-processed EEG signals, wherein the signals undergo band-pass filtering, followed by

segmentation and normalization. Thereafter, CCA technique is employed to maximize the correlation between the EEG signals and diagnostic measures associated with ADHD. The processed EEG signals enable extraction of discriminative features to distinguish between ADHD and non-ADHD populations efficiently and accurately. These signals are fed to the proposed CNN architecture for training and testing. The proposed methodology achieves an accuracy of 100% for classifying ADHD signals considering 19-channel EEG data. Further, we also focus on finding optimal channels that can reduce the computational load while achieving high accuracy. In this regard, it has been observed that the frontal brain EEG electrodes yield an accuracy of 99.08%. Since the research provides evidence regarding the significance of these channels for ADHD detection, the claim has been verified in this work to conclude that the frontal brain electrodes offer an efficient alternative as compared to using all 19 EEG channels.

In future, EEG-based ADHD detection models can be validated on diverse and large-scale datasets across different demographics, clinical settings, and geographic regions to ensure generalizability and reliability of the developed method. Moreover, we can collaborate with neurologists, psychiatrists, machine learning experts and data scientists to gain expert insights into interpreting EEG data and translate the research findings into clinical practice, ensuring that EEG-based ADHD detection tools are accurate, accessible, and compliant with regulatory standards. Further, the predictive capabilities of the model could be refined by integration of newer technologies and emerging machine learning techniques, such as transfer learning, ensemble methods and deep reinforcement learning. These advancements can lead to more accurate, reliable, and clinically applicable tools for ADHD diagnosis and monitoring, ultimately improving patient outcomes and advancing research in this field.

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Data availability

The data is publicly available at <https://ieee-dataport.org/open-access/eeeg-data-adhd-control-children>.

Declarations

Conflict of interest

The authors have no conflict of interest to declare that are relevant to the content of this article.

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